

2025 Ch H1 Q12

Section: Nature's Chemistry

Topic: Fragrances (Terpenes)

Question summary

Which of the following molecular formulae represents a terpene?

A. $C_{13}H_{20}$ B. $C_{15}H_{24}$ C. $C_{17}H_{28}$ D. $C_{19}H_{32}$

Worked solution

Terpenes are natural compounds built from isoprene units (C_5H_8).

The general formula for a terpene is $(C_5H_8)_n$, where n is an integer (2, 3, 4, ...).

Check each option:

$C_{13}H_{20}$ -> not a whole multiple of C_5H_8 .

$C_{15}H_{24}$ -> $3 \times (C_5H_8)$ -> valid terpene.

$C_{17}H_{28}$ -> not a multiple.

$C_{19}H_{32}$ -> not a multiple.

Therefore, the correct molecular formula for a terpene is $C_{15}H_{24}$.

Final answer

B. $C_{15}H_{24}$

Revision tips

- Terpenes are constructed from isoprene (C_5H_8) units linked head to tail.
- Common examples: limonene, myrcene, and pinene (all have formulas that are multiples of C_5H_8).
- They are found in essential oils and are responsible for the scents of many plants.